



Modelling Financial Markets

Chapter 0: Financial Markets in 2026



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Acronyms used in this material (1/3)

- AI: Artificial Intelligence
- AMM: Automated Market Maker
- APT: Arbitrage Pricing Theory
- AR: AutoRegressive (model); in event studies: Abnormal Return
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ASE: Academia de Studii Economice din București (the Bucharest University of Economic Studies)
- AUM: Assets Under Management
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BIS: Bank for International Settlements
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CAPM: Capital Asset Pricing Model
- CBOE: Chicago Board Options Exchange
- CCP: Central CounterParty
- CDO: Collateralised Debt Obligation
- CFTC: Commodity Futures Trading Commission
- CIR: Cox–Ingersoll–Ross (model)
- COVID-19: COronaVirus Disease 2019
- DJIA: Dow Jones Industrial Average
- EBA: European Banking Authority
- EMH: Efficient Market Hypothesis
- EOD: End of Day
- EODHD: EOD Historical Data (market-data provider)
- ES: Expected Shortfall

Acronyms used in this material (2/3)

- ESMA: European Securities and Markets Authority
- ETF: Exchange-Traded Fund
- EVT: Extreme Value Theory
- EWMA: Exponentially Weighted Moving Average
- FRED: Federal Reserve Economic Data
- FRTB: Fundamental Review of the Trading Book
- FX: Foreign eXchange
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GMM: Generalized Method of Moments
- GPT: Generative Pre-trained Transformer
- GPU: Graphics Processing Unit
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HFT: High-Frequency Trading
- HJM: Heath–Jarrow–Morton (model)
- HML: High Minus Low (value factor)
- IBM: International Business Machines (Corporation)
- IPO: Initial Public Offering
- LLM: Large Language Model
- MC: Monte Carlo (simulation)
- MDD: Maximum DrawDown
- MDE: Minimum Detectable Effect
- ML: Machine Learning
- MTF: Multilateral Trading Facility
- NISQ: Noisy Intermediate-Scale Quantum
- NYSE: New York Stock Exchange
- OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))

Acronyms used in this material (3/3)

- OTC: Over-The-Counter
- QAE: Quantum Amplitude Estimation
- QR: Quick Response (code)
- ROL: Romanian old leu (ISO 4217 code, until 2005)
- SD: Standard Deviation
- SE: Standard Error
- SEC: Securities and Exchange Commission
- SMB: Small Minus Big (size factor)
- SPDR: Standard & Poor's Depositary Receipts
- SR: Sharpe Ratio
- US: United States
- UTC: Coordinated Universal Time (Temps Universel Coordonné)
- VIX: CBOE Volatility Index
- VOC: Vereenigde Oostindische Compagnie (Dutch East India Company)

Chapter Route and Outcomes

Route of this chapter

- ▣ Part I: the financial system and market structure, a short history of exchanges and financial models, the course data, measurement and its precision
- ▣ Part II: the 2000–2026 landscape, the Romanian market, digital assets, passive investing and a case study on it, one pass of the AI-assisted research loop
- ▣ Technical detail sits in the [Appendix](#) at the end

After this chapter you can

- ▣ compute returns, CAGR, volatility, drawdowns and correlations on the correct calendar
- ▣ attach a standard error to a mean return, a Sharpe ratio and a correlation change
- ▣ explain how ETFs, stablecoins and market structure shape the data you model

Course Organisation

Programme and team

- ▣ Master's programme **Applied Statistics and Data Science**, Faculty of Cybernetics, Statistics and Economic Informatics, ASE
- ▣ 2 hours of lectures + 2 hours of seminars per week, 2026/2027
- ▣ Lecture: Prof. Daniel Traian Pele, danpele@ase.ro
- ▣ Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

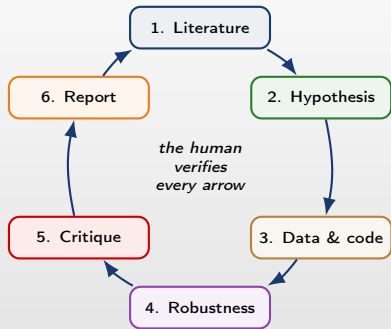
Grading

- ▣ Team project (GitHub, presentation, oral defence): **70%**
- ▣ Quizzes and seminar activity: **20%**; one quiz per chapter on the course website, 20 questions drawn from a bank, each answer explained
- ▣ Attendance: **10%**; at least **4 lectures** and **4 seminars**, recorded with the code of the day shown in the room (QR code or the Attendance form on the website)

The Team Project and Its Outputs

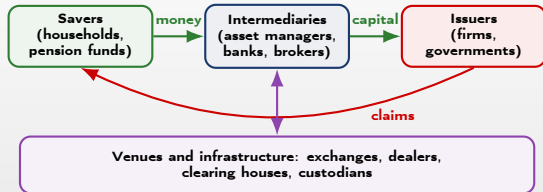
- ▣ Teams of 3–4 students, one research question on real market data; final presentation in [Chapter 19](#)
- ▣ One public GitHub repository per team
 - ▶ README (question, data, how to run), weekly commits that document the progress
 - ▶ every figure as a [Quantlet](#): one folder with code, `Metainfo.txt` (name, description, keywords, inputs, outputs) and the chart
- ▣ A result exists only if someone else can reproduce it from the repository
 - ▶ data loaded by code, every reported number produced by a script, random numbers with a fixed seed
- ▣ Tools: Python (pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib), notebooks in Google Colab, the course repository github.com/danpele/MFM
- ▣ AI use is allowed and declared: every use in [AI_USE.md](#), every caught AI error in [AI_ERRORS.md](#)

The AI-Assisted Research Loop



- An AI assistant (an LLM such as Claude, ChatGPT or Gemini) can draft every step: literature map, competing hypotheses, code, robustness plans, a hostile referee report
- The human owns the question, the identifying assumption, the test and every reported number
- Verification checklist
 - ▶ every reference: the DOI (Digital Object Identifier) resolves and the title matches
 - ▶ every number: reproduced from the data by your own code
 - ▶ H_0 , test, sample and multiple-testing correction fixed before seeing results
- Typical LLM failures: invented references, plausible but wrong formulas, look-ahead in code, searching until something is “significant”

What Markets Do and Who Trades



- **Financial market:** a mechanism where buyers and sellers exchange claims on future cash flows (shares, bonds, currencies, contracts, tokens) at prices set by supply and demand
- **Functions:** **capital allocation**, **risk sharing**, **liquidity** (converting assets to cash quickly), **price discovery** (prices aggregate dispersed information), **payments and settlement**
- **Buy side:** retail and institutional investors, asset managers, **passive** (replicate an index) or **active** (try to beat it)
- **Sell side:** brokers execute orders, dealers trade from inventory, **market makers** quote a buy and a sell price

Asset Classes

Class	Examples	Cash flow	Main risks
Equities	shares, equity ETFs	dividends, residual claim	market, firm-specific
Fixed income	government and corporate bonds	coupons, principal	interest rate, credit
Money market	T-bills (< 1 year), repo (secured loan)	short-term interest	liquidity
FX	EUR/USD, EUR/RON	interest differential	currency, policy
Commodities	gold, oil futures	none (roll yield)	supply, storage
Derivatives	futures, options, swaps	contingent payoffs	leverage, counterparty
Digital assets	Bitcoin, Ether, stablecoins	none / protocol-specific	volatility, regulation

- **ETF** (exchange-traded fund): a fund whose shares trade like a stock; **FX**: the currency market, EUR/RON = lei per euro; **roll yield**: gain or loss from replacing an expiring futures contract
- One statistical toolbox for all of them: returns, volatility, dependence and tails (full list of terms in the [Appendix](#))

Where Trading Happens

Venue	How trades are matched	What is visible	Examples
Exchange	central limit order book, auctions	quotes and trades	NYSE, Nasdaq, BVB
OTC dealer	bilateral, with a dealer	little, after the trade	bonds, FX, swaps
Dark pool	at a price taken from public markets	nothing before the trade	large equity orders
Centralised crypto exchange	order book run by a company, 24/7	quotes and trades	spot Bitcoin, Ether
AMM (decentralised)	a pool of two tokens, price set by a formula	the pool on the blockchain	Chapter 16

- **OTC**: over the counter, negotiated between two parties without an exchange; **AMM**: automated market maker, a smart contract that trades without an operator
- Crypto trades every day of the year, on hundreds of venues: [Makarov & Schoar \(2020\)](#) find large, persistent cross-exchange price gaps that capital controls and costs keep open

The Limit Order Book and a Market Order



- **Limit order book:** all waiting buy (**bid**) and sell (**ask**) orders, sorted by price; best bid 99.99, best ask 100.01
- **Bid-ask spread:** best ask minus best bid (0.02), the price of trading at once; **depth:** quantity near the best prices
- A **market buy** of 5 units consumes the book (orange)
 - ▶ 2 units at 100.01 and 3 at 100.02: average $(2 \times 100.01 + 3 \times 100.02)/5 = 100.016$
 - ▶ the best ask moves to 100.02 and the spread widens to 0.03: **market impact** ([Chapter 10](#))
- Order types: a resting **limit order** provides liquidity; a **market order** consumes it; a **stop order** becomes a market order at a trigger price, and clustered stops can accelerate a fall

Auctions versus Continuous Trading

Call auctions

- ▣ Orders are collected, then matched at the one price that maximises traded volume
- ▣ Used at the open and the close of most exchanges
- ▣ Concentrate liquidity: the closing auction sets the daily close used in this course

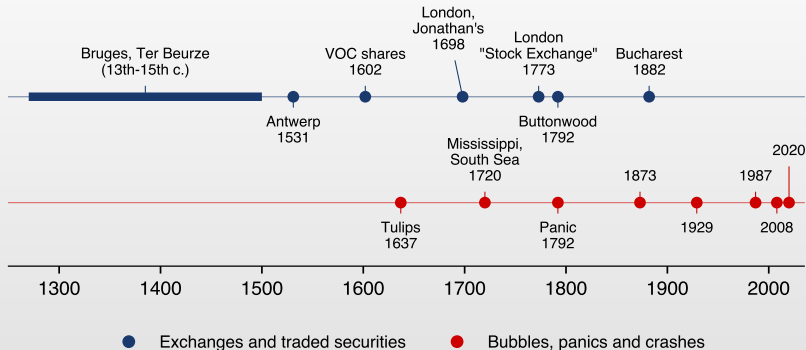
Continuous trading

- ▣ Each incoming order is matched at once if possible: best price first, then the earliest order
- ▣ Speed is rewarded: **high-frequency trading (HFT)** programs trade in microseconds
- ▣ Budish, Cramton & Shim (2015) propose frequent batch auctions to end the speed race

Observation Times and Price Noise

- A “daily price” is a price observed at a specific time
 - ▶ exchange-traded assets: the closing auction (S&P 500: 20:00 or 21:00 UTC)
 - ▶ Bitcoin: no close; the price of day t is taken at 24:00 UTC, the end of day t
 - ▶ EUR/RON: the BNR reference rate, fixed around 13:00 Bucharest time
- Different observation times split the same news across two days: correlations of assets with different closes are attenuated (▶ Appendix)
- **Bid–ask bounce**: trades alternate between bid and ask, so observed price changes are negatively autocorrelated even when the efficient price is a random walk
 - ▶ the Roll model gives $\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$ for a spread s (Roll (1984); derivation in the ▶ Appendix)
- Spreads, impact and fragmentation add noise to observed prices, most visible at high frequency (Chapters 9 and 10)

Five Centuries of Exchanges and Crashes



- Exchanges grew where merchants met regularly; shares traded continuously from the 1600s, and bubbles and crashes followed almost at once
- The Bucharest Stock Exchange (1882) has its own frames in the [Appendix](#)

The First Exchanges: from Bruges to New York



The Amsterdam exchange of Hendrick de Keyser, 1612

Engraving: Claes Jansz. Visscher (1612); public domain;

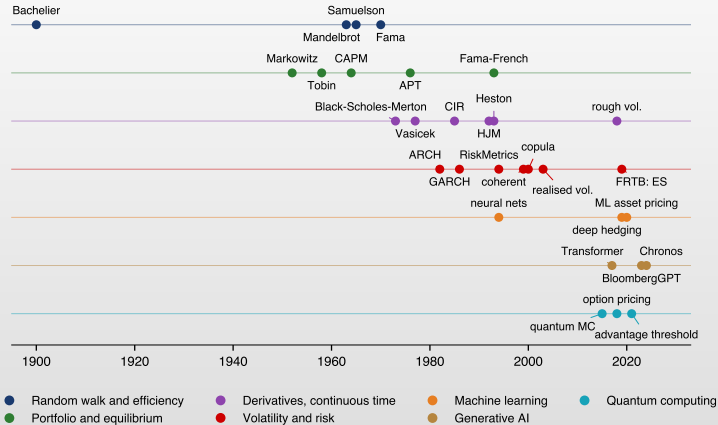
Stadsarchief Amsterdam, Wikimedia Commons

- **Bruges**, 13th–15th centuries: merchants met on the square before the inn of the Van der Beurse family; the name became *bourse*, *beurs*, *bursă* (Gelderblom (2013))
- **Antwerp**, 1531: a building purpose-built for daily trading opens (Gelderblom (2013))
- **Amsterdam**, 1602: the Dutch East India Company, VOC (Vereenigde Oostindische Compagnie), issues transferable shares
 - ▶ continuous trading, forward contracts, options and short selling within a decade (Gelderblom & Jonker (2004); Petram (2014))
- **London**: from 1698 John Castaing publishes price lists at Jonathan's Coffee House; in 1773 the brokers call their new building the Stock Exchange (Neal (1990))
- **New York**, 17 May 1792: 24 brokers sign the Buttonwood Agreement, the origin of the NYSE (New York Stock Exchange) (Sylla, Wright & Cowen (2009))

The First Bubbles, Panics and Crashes

- **Tulip mania**, Holland 1636–37: prices of bulb contracts rose many-fold, then collapsed in February 1637
 - ▶ revisionist evidence: rare bulbs lost value the way new flower varieties usually do ([Garber \(1989\)](#)); few bankruptcies and little damage to the economy ([Goldgar \(2007\)](#)); more in the [Appendix](#)
- **1720**: the Mississippi Company in Paris (John Law) and the South Sea Company in London rise and collapse within months
 - ▶ prices rose most for firms tied to the new Atlantic trade and to insurance: a bubble around a real innovation ([Frehen, Goetzmann & Rouwenhorst \(2013\)](#))
- **Panic of 1792**, New York: the Treasury under Alexander Hamilton acts as a lender of last resort ([Sylla, Wright & Cowen \(2009\)](#))
- **1873**: crashes in Vienna (May) and New York (September); the NYSE closes for ten days and the US contraction lasts until 1879 ([NBER business-cycle dates](#))
- **1929**: Black Tuesday, in the [Appendix](#)
- **19 October 1987**: the DJIA (Dow Jones Industrial Average) falls 22.6% in one day, with large falls in every major market ([Roll \(1988\)](#))
- Recurring ingredients: a new asset or technology, credit and leverage, then forced selling ([Chapter 17](#))

Financial Models, 1900–2026: an Overview



Origins: Random Walks and Efficient Markets



Louis Bachelier

Photo: unknown author; public
domain; Wikimedia Commons

- Bachelier (1900), *Théorie de la spéculation*: prices of French government bonds follow a Brownian motion, five years before Einstein; the first option-pricing formula
- Mandelbrot (1963): cotton price changes have far heavier tails than the Normal distribution allows
- Samuelson (1965): geometric Brownian motion keeps prices positive and makes log returns Normal
- Fama (1965); Fama (1970): the EMH (efficient market hypothesis), prices fully reflect available information
 - ▶ weak form (past prices), semi-strong form (public information), strong form (all information)
- In this course: stylised facts (Chapter 1), efficiency tests (Chapter 2), continuous-time models (Chapter 11)

Portfolios and Equilibrium Pricing



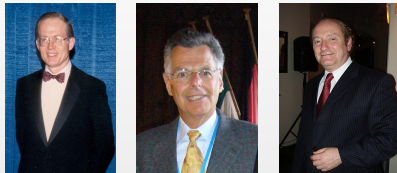
William F. Sharpe, 2007

Photo: Larry D. Moore (2007); CC

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- **Markowitz (1952)**: choose portfolio weights by mean and variance; diversification depends on covariances
- **Tobin (1958)**: with a riskless asset, every investor holds the same risky portfolio (two-fund separation)
- **CAPM (capital asset pricing model)**: $\mathbb{E}[R_i] - r_f = \beta_i (\mathbb{E}[R_m] - r_f)$, only market risk is priced (**Sharpe (1964)**; **Lintner (1965)**; **Mossin (1966)**)
- **APT (arbitrage pricing theory)**: no arbitrage with K common factors gives a linear factor model (**Ross (1976)**)
- Three factors: market, size (SMB, small minus big) and value (HML, high minus low book-to-market) (**Fama & French (1993)**)
- In this course: factor models (**Chapter 3**), portfolio optimisation (**Chapter 4**)

Derivatives and Continuous Time



Fischer Black, Myron Scholes, Robert C. Merton

Black: Dalmatine; public domain; Scholes:

Nobellaureatesphotographer (2008); CC BY-SA 3.0;

Merton: Digarnick (2006); public domain; Wikimedia

Commons

- Black & Scholes (1973); Merton (1973): a continuously rebalanced hedge replicates an option, so its price does not depend on the expected return of the stock
- Interest rates
 - ▶ Vasicek (1977): a mean-reverting short rate
 - ▶ CIR (Cox, Ingersoll & Ross (1985)): a square-root diffusion keeps the rate non-negative
 - ▶ HJM (Heath, Jarrow & Morton (1992)): the whole forward curve evolves without arbitrage
- Heston (1993): stochastic volatility with a closed-form option price
- Gatheral, Jaisson & Rosenbaum (2018): log-volatility behaves like a fractional Brownian motion with a Hurst exponent near 0.1 (rough volatility)
- In this course: Chapters 11 and 12

Volatility and Risk Measurement



Robert F. Engle, 2022

Photo: Jon Demske (2022); CC
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- ARCH (autoregressive conditional heteroskedasticity, [Engle \(1982\)](#)): today's variance depends on yesterday's squared shocks; GARCH (generalised ARCH, [Bollerslev \(1986\)](#)) adds the lagged variance
- RiskMetrics (J.P. Morgan, 1994): VaR (value at risk) from an EWMA (exponentially weighted moving average) of squared returns, $\lambda = 0.94$ for daily data ([J.P. Morgan/Reuters \(1996\)](#))
- 1996: banks may compute market-risk capital with internal VaR 1% models ([BCBS \(1996\)](#))
- [Artzner et al. \(1999\)](#): VaR is not a coherent risk measure, it can penalise diversification; ES (expected shortfall) is coherent ([Acerbi & Tasche \(2002\)](#))
- The FRTB (fundamental review of the trading book) replaces VaR 1% with ES 2.5% ([BCBS \(2019\)](#))
- EVT (extreme value theory) for the tails ([Embrechts, Klüppelberg & Mikosch \(1997\)](#); [McNeil & Frey \(2000\)](#)); realised volatility from intraday returns ([Andersen et al. \(2003\)](#))
- In this course: [Chapters 5–9](#)
- How the Gaussian copula mispriced the tranches of CDOs (collateralised debt obligations) before 2008: [► Appendix](#)

Machine Learning in Finance

- ML (machine learning): flexible functions fitted to data and selected by out-of-sample performance rather than by theory
- [Hutchinson, Lo & Poggio \(1994\)](#): neural networks learn option prices and hedge ratios from data, without the Black–Scholes formula
- [Gu, Kelly & Xiu \(2020\)](#): about 30,000 US stocks, 1957–2016, 94 firm characteristics
 - ▶ trees and neural networks beat linear models out of sample, through nonlinear interactions
 - ▶ the monthly out-of-sample R^2 stays below 1%: a small but economically valuable signal
- [Buehler et al. \(2019\)](#), deep hedging: a neural network chooses hedge positions under transaction costs to minimise a convex risk measure, with no pricing model
- Main risks: overfitting, look-ahead and many trials on the same data ([Chapters 13 and 14](#))

Generative AI and Foundation Models

- The Transformer ([Vaswani et al. \(2017\)](#)): attention layers read a whole sequence in parallel; it underlies every LLM (large language model) and GPT (generative pre-trained Transformer)
- BloombergGPT ([Wu et al. \(2023\)](#)): a 50-billion-parameter LLM trained on 363 billion tokens of financial text and 345 billion general tokens
- [Lopez-Lira & Tang \(2026\)](#): ChatGPT scores of news headlines predict next-day stock returns out of sample
- Foundation models for time series: Chronos scales and quantises values into tokens and trains a language model on many series, then forecasts new series zero-shot ([Ansari et al. \(2024\)](#))
- Open problems
 - ▶ look-ahead: the model may have read texts written after the forecast date
 - ▶ cost, reproducibility and the decay of a signal once many traders use it
- In this course: [Chapters 14 and 15](#)

Quantum Computing: Promise and Limits



IBM Quantum System One, Ehningen
(Germany), 2021

Photo: IBM Research (2021); CC BY 2.0;
[Wikimedia Commons](#)

- Qubits hold superpositions, and interference amplifies the right answer (Orús, Muga & Lizaso (2019); Herman et al. (2023))
- **Proven:** to estimate a mean (an option price, a VaR) with error ε , classical MC (Monte Carlo) needs about $1/\varepsilon^2$ samples, QAE (quantum amplitude estimation) about $1/\varepsilon$ steps (Montanaro (2015))
 - ▶ $\varepsilon = 10^{-4}$: 10^8 samples against 10^4 steps; the gain is *quadratic*: the number of steps becomes the square root of the classical one, it does not fall exponentially
- **Demonstrated** on a few qubits: option pricing (Rebentrost, Gupta & Bromley (2018); Stamatopoulos et al. (2020)), VaR and ES (Woerner & Egger (2019))
- **Not yet practical**
 - ▶ today's processors are NISQ (Noisy Intermediate-Scale Quantum): tens to a few hundred qubits without error correction, so noise spoils the result after a few operations (Preskill (2018))
 - ▶ loading a return distribution into qubits is costly
 - ▶ a useful advantage in derivative pricing needs about 8,000 error-corrected qubits, machines that do not exist yet (Chakrabarti et al. (2021))
- The calculation step by step: [▶ Appendix](#)

What Changed Each Time, and Where It Appears in This Course

Wave	What made it possible	Chapters
Random walk, efficiency	probability theory, long price series	1, 2, 11
Portfolio, equilibrium	utility theory, the first stock-return databases	3, 4
Derivatives, continuous time	stochastic calculus, listed options (1973)	11, 12
Volatility and risk	daily and intraday data, bank regulation	5–9
Machine learning	large panels, cheap computing	13
Generative AI	text at scale, GPU (graphics processing unit)	14, 15
Quantum computing	hardware, the Transformer new hardware, not yet practical	outlook

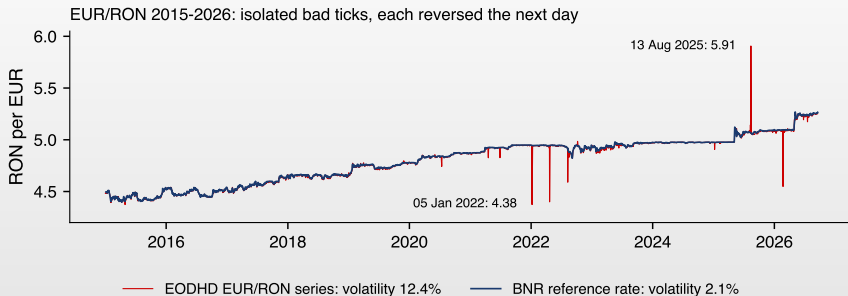
- No wave replaced the earlier ones: GARCH still feeds VaR, factor models still benchmark ML, Black–Scholes still quotes implied volatility
- The complete course map is in the [Appendix](#)

Data Used in This Course

Data	Content	Format
Daily market data from EODHD (EOD Historical Data), saved in the course repository	indices, stocks, ETFs, FX, crypto, yields	CSV
FRED (Federal Reserve Economic Data)	US rates, exchange rates, macro series	public CSV
BNR	official RON reference exchange rates	public XML
DefiLlama	stablecoin supply, DeFi (decentralised finance)	public JSON

- ▣ Stocks and ETFs: adjusted close (dividends, splits); indices, FX, crypto and yields: close
- ▣ One series: returns on its own calendar; joint analysis: join the *prices* on common days, then take returns (Friday → Monday for both)
- ▣ Annualise with the observed frequency: 365 for crypto, about 260 for spot gold on weekdays, 252 for equities
- ▣ **Survivorship**: samples rebuilt from today's survivors, without delisted firms or dead coins, make past returns look too good ([Brown, Goetzmann, Ibbotson & Ross \(1992\)](#))

Data Pitfalls: a Plotted Bad Tick



- ❑ **Bad tick:** a wrong price, a spike reversed the next day; EUR/RON series from EODHD (aggregated interbank quotes, not an official rate): $\pm 15\%$ on 13–14 Aug 2025, $\pm 12\%$ in early Jan 2022
- ❑ Annualised volatility 2015–2026: 12.4% with these ticks, 2.1% with the BNR reference rate; EUR/RON always from BNR
- ❑ Romania 10-year yield: a 0.00% print on 24 Dec 2013 and stale values on Sundays in late 2022

Returns, Growth and Annualisation

- ▣ **Price index** (S&P 500, BET) ignores dividends; **total-return index** (BET-TR) and **adjusted close** (stocks, ETFs) reinvest them
- ▣ Simple return $R_t = P_t/P_{t-1} - 1$; log return $r_t = \ln(P_t/P_{t-1}) = \ln(1 + R_t)$, additive over time
- ▣ **CAGR** (compound annual growth rate) $= (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$, with $Y = \text{calendar days}/365.25$
- ▣ **Annualised volatility** $\hat{\sigma}_d\sqrt{q}$: $\hat{\sigma}_d$ = standard deviation of daily log returns, q = observed return intervals per year (about 252 for equities, 365 for crypto)
- ▣ Worked example: S&P 500, 2 Jan 2015 – 18 Sep 2026 (4 277 days, $Y = 11.71$)
 - ▶ multiple $\times 3.72$: $\text{CAGR} = 3.72^{1/11.71} - 1 = e^{\ln 3.72/11.71} - 1 = 11.9\%$
 - ▶ with dividends reinvested (SPY, the SPDR S&P 500 ETF): 13.7% a year; the 1.8 percentage point (pp) gap is the dividend yield

Risk, Dependence and Rates: Notation

- **Drawdown** $DD_t = P_t / \max_{s \leq t} P_s - 1$; **maximum drawdown (MDD)** $= \min_t DD_t$, the worst loss from a previous peak
- **Correlation**: Pearson correlation of returns, never of prices; **rolling**: on a moving 252-day window
 - ▶ two calendars (stocks vs crypto): join the *prices* on common days first, then compute returns
- **Yield to maturity**: the internal rate of return of a bond's promised cash flows
 - ▶ equals the realised return only if the bond is held to maturity, pays in full and coupons are reinvested at the same yield
- **Spread** $y^{10y} - y^{2y}$ in percentage points (pp); 1 basis point (bp) $= 0.01$ pp; **inversion**: spread < 0

How Precisely Do We Know an Expected Return?

- Daily returns independent and identically distributed (iid), with mean μ_d and standard deviation σ_d ; $T = qY$ observations over Y years
- Annualised mean $\hat{\mu} = q\bar{R}$ has standard error

$$\text{SE}(\hat{\mu}) = \frac{q\sigma_d}{\sqrt{T}} = \frac{\sigma_a}{\sqrt{Y}}, \quad \sigma_a = \sigma_d\sqrt{q}$$

- It depends only on the **span** Y , not on the sampling frequency ([Merton \(1980\)](#))
- S&P 500 with dividends (SPY), 2015–2026: $\hat{\mu} = 14.4\%$, $\sigma_a = 17.5\%$, $Y = 11.71$
 - ▶ $\text{SE} = 17.5\%/\sqrt{11.71} = 5.1 \text{ pp}$; 95% confidence interval [4.3%, 24.4%]
 - ▶ halving the SE needs four times the span (47 years), not more frequent data

$\sqrt{h}$ Scaling and the Variance Ratio

- Stationary returns with variance σ^2 and autocorrelations ρ_j :

$$\text{Var} \left(\sum_{j=1}^h r_{t+j} \right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} \left(1 - \frac{j}{h} \right) \rho_j \right]$$

- The $\sqrt{h}$ rule is exact when returns are **uncorrelated** with **constant variance**; independence is not needed
- Variance ratio** $\text{VR}(h) = \text{Var}(h\text{-day return})/(h\sigma^2)$; for an AR(1), $\text{VR}(h) \rightarrow (1 + \rho)/(1 - \rho)$
- S&P 500, 2000–2026: $\hat{\rho}_1 = -0.10$ and $\widehat{\text{VR}}(20) = 0.75$
 - scaling daily volatility by $\sqrt{20}$ overstates the 20-day (monthly) risk
 - use the whole autocorrelation function or aggregate the returns directly (more examples in the [Appendix](#); tests in [Chapter 2](#))

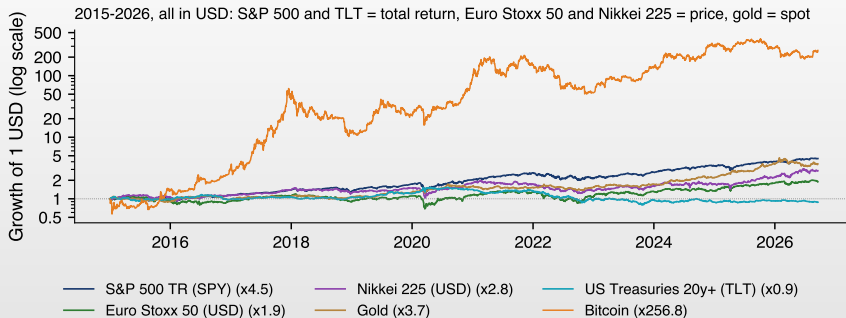
The Sharpe Ratio Is an Estimate

- ▣ **Sharpe ratio (SR)**, risk-free rate set to 0: $SR_d = \mu_d / \sigma_d$ per day, annualised $SR_a = \sqrt{q} SR_d$
- ▣ Delta method, iid returns following the Normal distribution (Lo (2002)):

$$\sqrt{T} (\widehat{SR}_d - SR_d) \xrightarrow{d} N(0, 1 + SR_d^2/2) \Rightarrow SE(\widehat{SR}_a) = \sqrt{\frac{1 + SR_a^2/(2q)}{Y}} \approx \frac{1}{\sqrt{Y}}$$

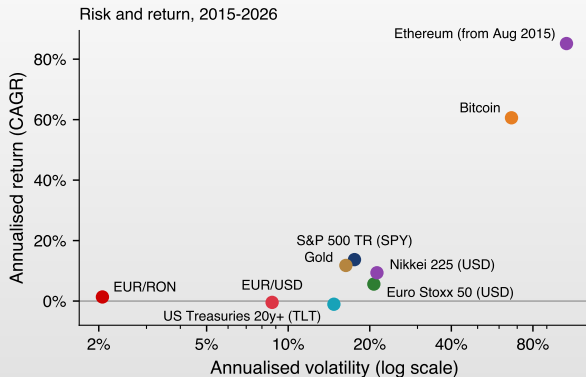
- ▣ SPY, 2015–2026: $\widehat{SR}_a = 0.82$, $SE = 0.29$, 95% confidence interval [0.25, 1.39]
- ▣ Like the mean, the SR is known only to about $\pm 2/\sqrt{Y}$: 11.7 years cannot separate 0.8 from 0.5
- ▣ Fat tails, dependence and picking the best of many strategies change the SE: [► Appendix](#)

Asset Classes since 2015



- Value of 1 USD invested on 2 Jan 2015, log scale; Euro Stoxx 50 and Nikkei 225 converted with daily [FRED](#) exchange rates
- SPY and TLT (iShares 20+ Year Treasury Bond ETF) are total return; Euro Stoxx 50 and Nikkei 225 are price indices, without dividends
- Long Treasuries ended below their 2015 level ($\times 0.9$); Bitcoin's $\times 257$ is one coin chosen ex post among thousands

Risk and Return, 2015–2026

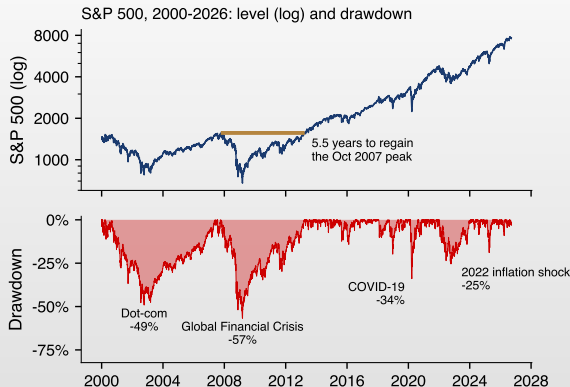


- Vertical axis: CAGR; horizontal: annualised volatility of daily log returns, with the observed q
- All in USD, except EUR/RON (RON per EUR, volatility 2.1% a year, a managed currency)
- S&P 500 with dividends: 13.7% a year at 17.6% volatility; gold 11.8% at 16.3% (weekday quotes)
- Bitcoin 60.6% at 66.7%; Ethereum, from Aug 2015 only, 85.2% at 106%
- Question: is any ranking of these points statistically defensible? (next slide)

Is Any Ranking in the Risk–Return Chart Significant?

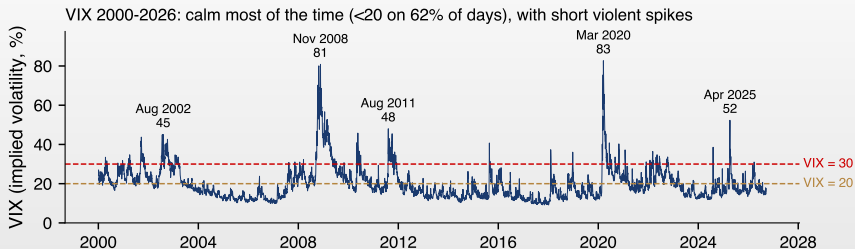
- Standard error of an annualised mean return, $\sigma_a/\sqrt{Y}$ with $Y = 11.7$
 - ▶ S&P 500 with dividends 5.1 pp, gold 4.8 pp, Euro Stoxx 50 6.0 pp, Nikkei 225 6.2 pp, Bitcoin 19.4 pp
 - ▶ S&P 500 minus gold on common days: +1.9 pp a year, SE 6.8 pp, $t = 0.28$
- Sharpe ratios: Bitcoin 1.05, S&P 500 0.82, gold 0.77, Nikkei 225 0.53, Euro Stoxx 50 0.37, TLT 0.00, each with SE ≈ 0.29
- Overlapping intervals do not test a difference: bootstrap the paired difference on common days (Seminar 0, B1 Extended)
 - ▶ S&P 500 minus gold: +0.06, 95% interval $[-0.65, +0.81]$
 - ▶ S&P 500 minus Euro Stoxx 50: +0.46, $[0.09, 0.85]$, significant alone; none of the 10 pairs of non-crypto assets survives a Bonferroni correction (each test at 0.5%)
- The chart describes the sample; it does not rank expected returns

Crises and Drawdowns



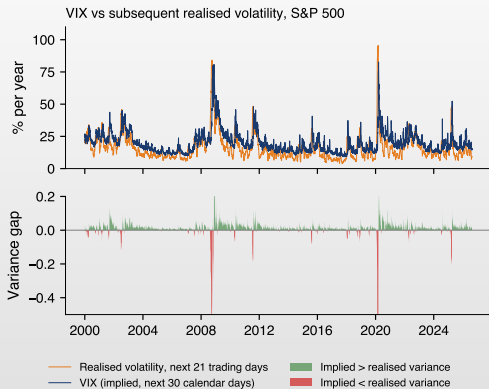
- Lower panel: $DD_t = P_t / \max_{s \leq t} P_s - 1$ of the S&P 500 (price index)
- Four troughs since 2000: dot-com -49% (Oct 2002), global financial crisis -57% (Mar 2009), COVID-19 -34% (Mar 2020, the fastest), inflation shock -25% (Oct 2022)
- 1 565.15 (9 Oct 2007) to 676.53 (9 Mar 2009); the peak was regained only on 28 Mar 2013
- The MDD deepens with the sample length: compare MDDs only over the same horizon
 ([Appendix](#))

Volatility Regimes: the VIX



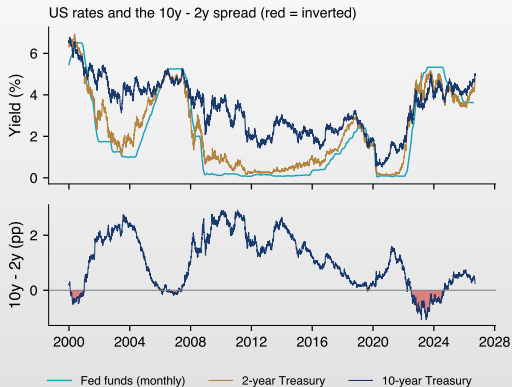
- ▣ **VIX** (CBOE Volatility Index): the S&P 500 volatility over the next 30 days **implied** by option prices, in % per year
- ▣ Since 2000: below 20 on 62% of days, above 30 on 9%; peaks 82.7 (16 Mar 2020), 80.9 (20 Nov 2008), 52.3 (8 Apr 2025)
- ▣ Long calm periods, short intense stress: volatility comes in regimes ([Chapters 5 and 9](#))

Implied versus Realised Variance



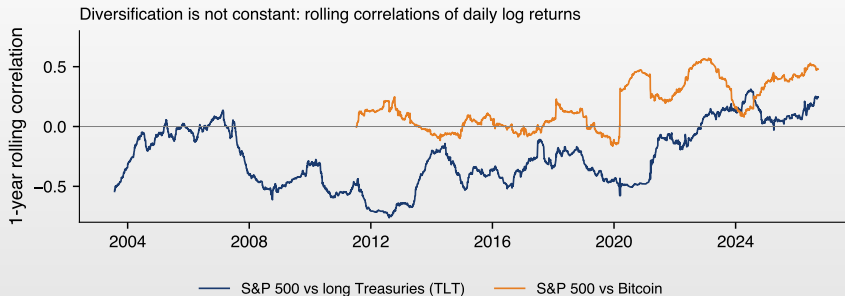
- $(VIX/100)^2 \approx$ the **risk-neutral** expectation of the annualised S&P 500 variance over the next 30 days
- **Variance risk premium:** risk-neutral minus real-world expected variance, positive on average ([Carr & Wu \(2009\)](#))
- Proxy here: realised volatility over the next 21 trading days, $\sqrt{(252/21) \sum r^2}$; incomplete final windows omitted
- 2000–2026: mean VIX 19.8 vs realised 16.2; the VIX was higher on 83% of days
- Ex post the gap is premium plus forecast error; [Bollerslev, Tauchen & Zhou \(2009\)](#) show that a proxy for the variance risk premium, known at t , predicts quarterly equity returns

Interest Rates and the Yield Curve



- **FRED**: 10-year and 2-year US Treasury yields; **fed funds rate**, the overnight rate targeted by the US central bank
- Spread 10y – 2y: briefly negative on 1–4 Apr 2022, then inverted from 6 Jul 2022 to 26 Aug 2024 (537 trading days), minimum -1.08 pp (3 Jul 2023)
- 18 Sep 2026: 10-year 5.01%, 2-year 4.76%, spread $+0.25$ pp = 25 bp; fed funds 3.63% (Aug 2026)
- A yield is not a realised return: TLT lost value in 2015–2026 as yields rose

Diversification Benefits Change over Time



- 252-day rolling correlations of daily log returns, prices joined on common days first
- Stocks vs long Treasuries: mean rolling -0.43 (2010–2020), $+0.08$ (2022–2026); whole-period correlations -0.47 and $+0.11$, a different estimator
- Stocks vs Bitcoin: mean rolling 0.01 (2016–2019), 0.38 (2022–2026); fixed-correlation portfolio models miss such shifts (Chapters 4 and 6)

Has the Stock–Bond Correlation Really Changed?

- ▣ Whole-period correlation S&P 500 – TLT: -0.47 in 2010–2020 ($n = 2\,769$), $+0.11$ in 2022–2026 ($n = 1\,182$); estimated change $+0.58$
- ▣ Fisher z test, iid pairs from a bivariate Normal distribution: $\operatorname{atanh} \hat{\rho}$ has variance about $1/(n - 3)$
 - ▶ $z = 17.9$, $p < 10^{-10}$; iid 95% interval for the change $[0.52, 0.65]$
- ▣ Volatility clustering makes pairs dependent and fat tails break normality: stationary bootstrap, mean block 20 days
 - ▶ 95% interval $[0.49, 0.68]$, 1.54 times wider; the change survives (Seminar 0, B3 Extended)
- ▣ The test establishes a change in association between two periods, not its cause

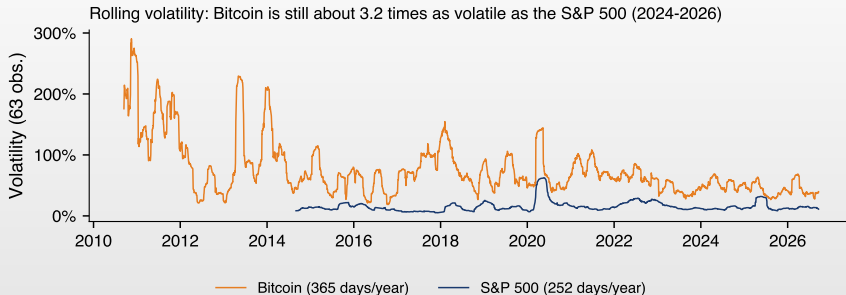
 MFM_ch0_correlations

Two Traps for Crypto–Equity Correlations

- **Asynchronous closes:** Bitcoin's day ends at 24:00 UTC, the S&P 500's 3–4 hours earlier, so part of the co-movement appears at lags
 - ▶ Bitcoin – S&P 500, 2022–2026: same day 0.42; S&P 500 with Bitcoin one day earlier +0.03, one day later –0.05; sum 0.40
 - ▶ diagnostic: the asynchronous-close bias is small for this pair at the daily frequency
- **Volatility bias:** correlations estimated on turbulent days are larger even when the link is stable ([Forbes & Rigobon \(2002\)](#))
 - ▶ VIX in its top quartile (> 21.8): 0.52 vs 0.32 on the other days
 - ▶ adjusted for the 4.64 times higher S&P 500 variance: 0.27, so the “stress correlation” is mostly a volatility effect
- Models, assumptions and correction formulas: [▶ Appendix](#)

 MFM_ch0_correlations

Bitcoin Remains Several Times as Volatile as Equities



- Rolling annualised volatility over the last 63 observations: about 3 months for the S&P 500 (annualised with $\sqrt{252}$), about 2 months for Bitcoin (annualised with $\sqrt{365}$)
- Bitcoin: 65% on average in 2015–2017, 46% in 2024–2026; S&P 500 in 2024–2026: 14%, so Bitcoin is still about **3.2 times** as volatile
- Volatility changes over time for both: a constant-volatility model is a poor description

The Bucharest Stock Exchange Today

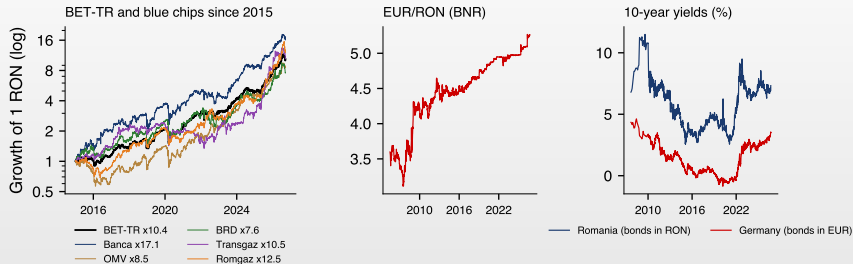
- **BVB** (Bursa de Valori București): opened in 1882, closed in 1948, re-established in 1995 (history in the [Appendix](#))
- **BET** (Bucharest Exchange Trading index): the most traded Romanian companies, launched on 19 Sep 1997 at 1 000 points [BVB \(indices\)](#); **BET-TR**: its total-return version, with reinvested dividends
- Largest companies: banks (Banca Transilvania, BRD), energy (OMV Petrom, Romgaz, Hidroelectrica), utilities (Transgaz)
- Classified as **Secondary Emerging** by [FTSE Russell \(2026\)](#) since September 2020
- Milestones: Hidroelectrica, the largest IPO (initial public offering) in BVB history, EUR 1.9 bn, July 2023 [BVB \(history\)](#); **TVBETETF**, an ETF on BVB that replicates BET

The BET Index, 1997–2026



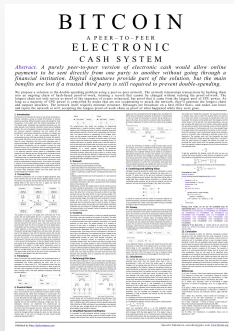
- Official closing values, log scale; peak 10 813.59 (24 Jul 2007), trough 1 887.14 (25 Feb 2009): −82.5%, regained only on 16 Mar 2021, after 13.6 years
- 19 Dec 2018: −11.2% in one day, the day after the bank-tax announcement; Mar 2020: −31.1% from the February high
- 18 Sep 2026: 32 757.81, ×32.8 since launch, CAGR 12.8% over 29 years (record 36 487.21 on 19 Aug 2026)

Romanian Equities, the Leu and Government Yields



- Dividend-adjusted growth since Jan 2015: BET-TR $\times 10.4$, Banca Transilvania $\times 17.1$; EUR/RON (BNR) from 3.60 (4 Jul 2005, after the redenomination 1 RON = 10 000 ROL) to 5.26
- 10-year yields, 18 Sep 2026: Romania 7.33% (bonds in RON), Germany 3.53% (in EUR): a cross-currency differential of 3.8 pp
- This differential mixes expected leu depreciation, inflation and policy-rate gaps with credit, liquidity and term premia: it is **not** a pure sovereign-credit spread

Bitcoin, Ethereum and Tokens



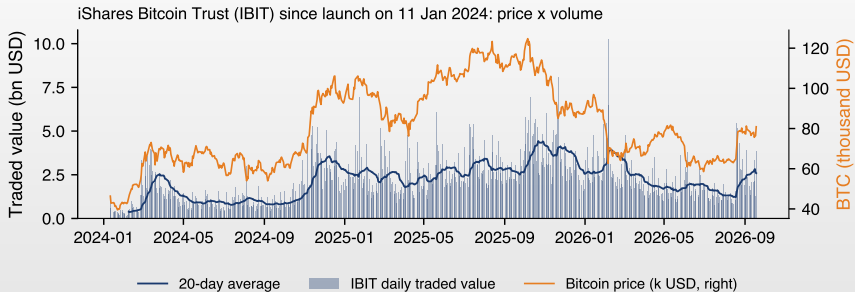
Bitcoin white paper (2008)

Satoshi Nakamoto (2008); CC0; Wikimedia

[Commons](#)

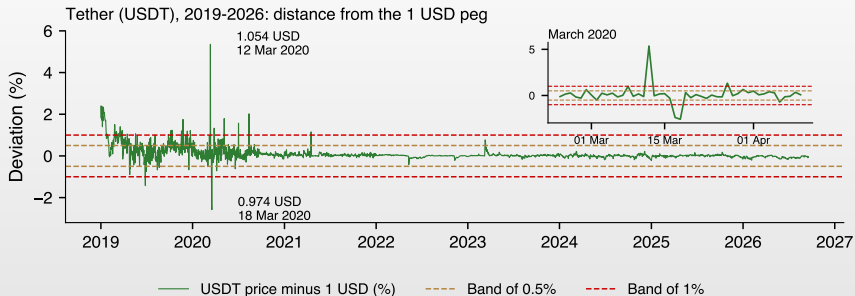
- 31 Oct 2008: *Bitcoin: A Peer-to-Peer Electronic Cash System*, by the pseudonymous Satoshi Nakamoto [Nakamoto \(2008\)](#)
- **Blockchain**: a shared ledger of transactions, copied on many computers and extended in blocks, without a trusted intermediary
- **Bitcoin** (2009): fixed supply rule (at most 21 million coins); **proof of work**: adding a block costs computing power, which makes rewriting history expensive
- **Ethereum**: a blockchain that runs programs (**smart contracts**); its coin is Ether; a **token** is a unit on a blockchain representing a coin, a claim or a right
- [Härdle, Harvey & Reule \(2020\)](#): statistical properties, pricing and regulation; [Pele et al. \(2023\)](#): tail, memory and moment factors set cryptos apart from stocks, FX, commodities, bonds and real estate

Spot Bitcoin ETFs



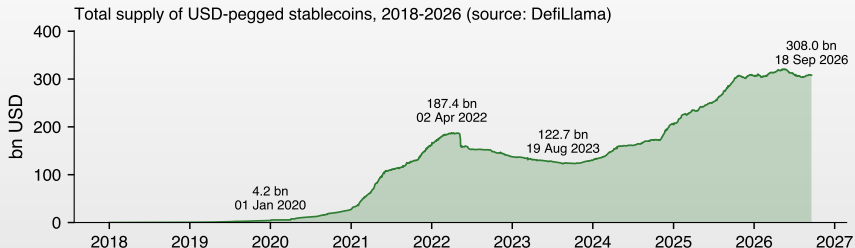
- **Spot** ETF: holds the asset itself, not futures; **ETP** (exchange-traded product): the wider category; spot Bitcoin ETPs approved in the US on 10 Jan 2024 ([SEC \(2024\)](#))
- IBIT (iShares Bitcoin Trust), trading since 11 Jan 2024: daily traded value = price $\times$ shares traded; median 1.97 bn USD a day, peak 10.3 bn USD (5 Feb 2026)

Stablecoins and the Peg



- **Stablecoin**: a token designed to keep a fixed value (**peg**), usually 1 USD; **reserve-backed** (cash, T-bills, redeemable at 1 USD) or **algorithmic** (a second token and trading rules); **depeg**: the price leaves the peg
- Tether (USDT), 2019–2026: within $\pm 0.5\%$ of 1 USD on 93% of days, within $\pm 1\%$ on 98%; extremes 0.974 and 1.054 USD in March 2020
- The algorithmic stablecoin TerraUSD lost its peg and collapsed in May 2022

Stablecoin Supply



- **Supply:** the USD value of USD-pegged stablecoins in circulation ([DefiLlama](#)): 4.2 bn (1 Jan 2020), peak 187.4 bn (2 Apr 2022), trough 122.7 bn (19 Aug 2023), 308.0 bn (18 Sep 2026)
- The 2022–2023 fall followed the TerraUSD collapse and the crypto crash; circulation value is distinct from the peg: supply can fall while the price stays at 1 USD

Tokenisation and Regulation

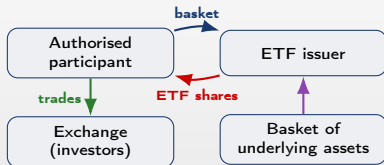
Tokenised real-world assets

- **RWA** (real-world assets): bonds, fund shares, deposits, commodities
 - A token on a shared ledger is a claim on such an asset: transfer and settlement become programmable and near-instant
 - Examples: tokenised money-market and Treasury funds
 - **BIS (2023)**: a “unified ledger”; **BIS (2025)**: stablecoins fail the tests for money (▶ [Appendix](#))
- Statistical question: do tokenised assets inherit the properties of the underlying, or those of crypto markets?

Regulation, key dates

- EU: **MiCA, Regulation (EU) 2023/1114** (Markets in Crypto-Assets)
- US: spot Bitcoin ETPs approved on 10 Jan 2024 (**SEC (2024)**); payment stablecoins under the **GENIUS Act, Public Law 119-27**
- Regulatory dates are natural break points and event-study dates

ETFs: Creation and Redemption



- **ETF**: an investment fund whose shares trade on an exchange like a stock; most are **passive** and replicate an index at low cost
- **Net asset value (NAV)**: the fund's assets minus its liabilities, per share, computed once a day
- **Authorised participant (AP)**: a large dealer allowed to deal directly with the issuer
- **Creation**: the AP delivers the basket and receives new ETF shares; **redemption**: the reverse
- This two-way channel keeps the market price close to the NAV; ETFs now hold equities, bonds, commodities and, in the US since January 2024, spot Bitcoin

Worked Example: Arbitrage Keeps the ETF Price Near NAV

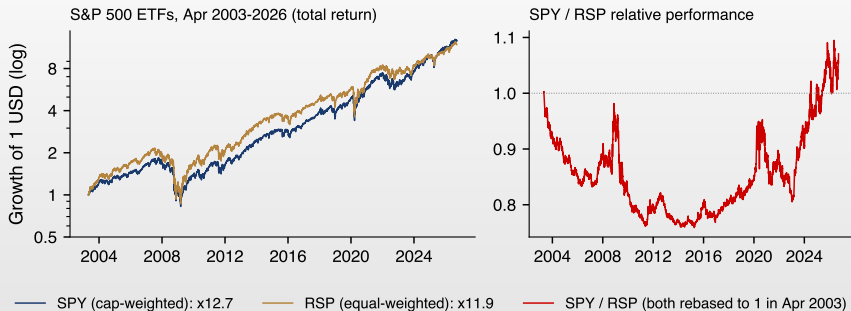
- ▣ Illustrative numbers: $NAV = 100$, the ETF trades at 100.50; **premium** $(P - NAV)/NAV = 0.5\%$
- ▣ The AP buys the basket for 100, delivers it to the issuer, receives one new ETF share and sells it for 100.50
 - ▶ profit 0.50 per share before costs; the extra ETF shares push the price down until the premium disappears
 - ▶ at a **discount** ($P < NAV$) the AP buys ETF shares and redeems them for the basket
- ▣ Premiums persist where the NAV is stale or the basket is illiquid, e.g. foreign stocks or bonds (Petajisto (2017))
- ▣ The speed at which premiums decay can be measured with an AR(1) half-life (▶ Appendix)

Cap-Weighted versus Equal-Weighted Indices

Definition 1 (Index weights and effective diversification)

- Capitalisation weights $w_i = MC_i / \sum_{j=1}^n MC_j$, market capitalisation $MC_i = \text{price} \times \text{number of shares}$; equal weights $w_i = 1/n$, rebalanced periodically
- Effective number of stocks $N_{\text{eff}} = 1 / \sum_i w_i^2$ (inverse Herfindahl): n for equal weights, far fewer when a few firms dominate
- SPY** (SPDR S&P 500 ETF): the S&P 500 with capitalisation weights; **RSP** (Invesco S&P 500 Equal Weight ETF): the same stocks, each $1/500$
- A cap-weighted index is less diversified than its number of stocks suggests; its risk depends on a few firms and themes ([Chapters 3, 4 and 17](#))
- Passive cap-weighted funds hold each firm in proportion to its size, so their investors carry the concentration of the largest firms

Market Concentration: SPY versus RSP



- Total return (adjusted close) since RSP's launch in April 2003: SPY $\times 12.7$, RSP $\times 11.9$; the ratio ends at 1.07
- SPY/RSP rose by 31% since January 2023: a handful of very large companies drove the index
- The ratio measures relative performance; it is not a direct measure of concentration or of passive ownership

Case Study: What Happens When Passive Investing Grows?

- ▣ Haddad, Huebner & Loualiche (2025), *American Economic Review* 115(3), 975–1018
- ▣ **Question:** when some investors turn passive, do the others trade more aggressively and undo the effect on prices?
- ▣ **Demand elasticity** $\mathcal{E}_i$: the percentage fall in the shares investor i demands after a 1% price rise; a passive investor has $\mathcal{E}_i = 0$
- ▣ **Aggregate elasticity** $\mathcal{E}_{agg}$: the holdings-weighted average of the $\mathcal{E}_i$ (eq. 2); a demand shift $\Delta \bar{d}$ moves the log price by $\Delta p = \mathcal{E}_{agg}^{-1} \Delta \bar{d}$ (eq. 3)
 - ▶ lower $\mathcal{E}_{agg}$: prices more volatile and less informative (Section II.A)
- ▣ Link: stocks with higher ETF ownership are more volatile (Ben-David, Franzoni & Moussawi (2018))

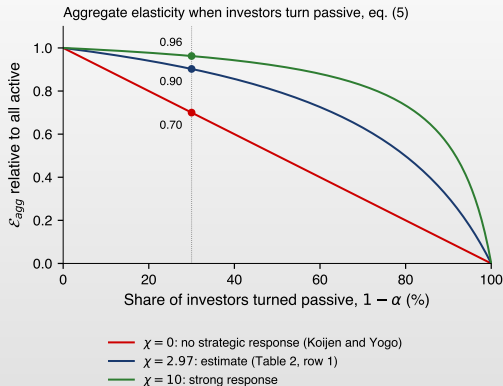
Case Study: How the Elasticity Is Estimated

- **Strategic response** χ : $\mathcal{E}_i = \bar{\mathcal{E}}_i - \chi \mathcal{E}_{agg}$ (eq. 4), with $\bar{\mathcal{E}}_i$ the investor's own component
 - ▶ $\chi = 0$: nobody reacts, the demand system of [Kojien & Yogo \(2019\)](#); $\chi \rightarrow \infty$: the others fully offset any change
- **Data**: quarterly holdings of US institutional managers (SEC Form 13F), 2001–2020, about 80% of US market capitalisation
- **Identification**: the same investor meets a different mix of other investors in each stock, which separates $\bar{\mathcal{E}}_i$ from χ (Theorem 1)
 - ▶ instruments: the money that would flow to each stock if the other investors held equal weights in their usual universes
- **Estimation**: GMM (generalised method of moments) with a nested equilibrium; the full demand system, equilibrium conditions and instruments are in the [Appendix](#)

Case Study: Headline Results

- Strategic response $\chi = 2.97$ (SE 0.47; Table 2, row 1): investors react, far from both 0 and ∞
- Demand for single stocks is inelastic: mean $\mathcal{E}_{agg} = 0.438$ (Table 3)
- **Pass-through** of a fall in the active share to $\mathcal{E}_{agg}$ (eq. 28): $1/(1 + 3 \times 0.68) = 33\%$
 - ▶ median active share 81.4% (2001) to 59.5% (2020): a fall of $\ln(81/59) = 32\%$
 - ▶ times the pass-through: stocks are about 11% less elastic; two-thirds of the direct effect is offset
- Economic implication: inelastic stocks are more volatile and less liquid (Table 5); robustness and reduced-form checks in the [Appendix](#)

Case Study: Strategic Response Damps the Effect of Passive Investing

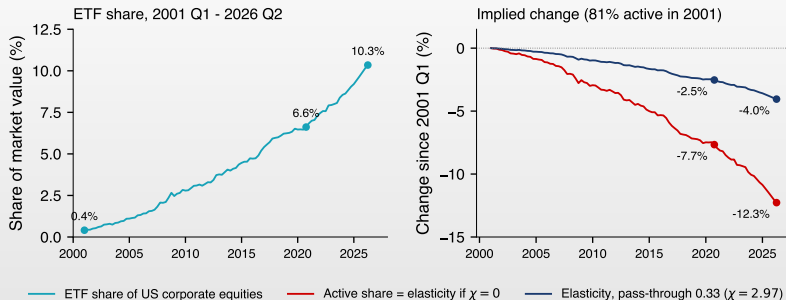


- Eq. (5): $\mathcal{E}_{agg}$ after a share $1 - \alpha$ of identical investors turns passive, relative to all active:

$$\alpha + (1 - \alpha) \frac{\chi \alpha}{1 + \chi \alpha}$$

- Worked point, 30% passive: $\chi = 0$ gives 0.70; $\chi = 2.97$ gives 0.90, two-thirds of the direct fall offset
- $\chi = 10$ still gives 0.96: only an infinite response keeps $\mathcal{E}_{agg}$ unchanged
- All curves reach 0 at 100% passive: nobody is left to absorb demand shocks

Case Study: How Much Elasticity Falls as the ETF Share Grows



- ETF share of US corporate equities (Financial Accounts of the United States): 0.4% (2001 Q1), 6.6% (2020 Q4), 10.3% (2026 Q2)
- Rule of Section V.A (81% active in 2001, pass-through 0.33): active share -7.7% and elasticity -2.5% to 2020; -12.3% and -4.0% to 2026 Q2
- A scenario, not a replication: ETFs include active funds and exclude index mutual funds

Case Study: Limits and an Open Question

- ▣ Limits stated by the authors
 - ▶ one χ for all investors; no strategic entry, exit or fund flows (Section I.B)
 - ▶ one 13F form per institution: its active and passive funds cannot be separated (footnote 51)
 - ▶ the elasticity is for single stocks, not for the market as a whole (footnote 37)
 - ▶ those who turn passive are assumed representative of active investors (Section V.A)
- ▣ Extension: does the pass-through change if the most elastic investors turn passive first? (Seminar 0, C3)
- ▣ Open question: does passive investing change price formation?
 - ▶ [Sammon \(2025\)](#): higher passive ownership makes prices anticipate earnings announcements less
 - ▶ passive ownership is endogenous: identification relies on index reconstitutions and flow shocks

Hypothesis: Is There a Concentration Premium?

- ▣ A typical assistant proposal: “passive flows favour the largest firms, so cap weights beat equal weights”
- ▣ Discovery loop in one pass
 - ▶ **literature**: ask for peer-reviewed papers with DOIs, then resolve each DOI yourself
 - ▶ **hypothesis**: three competing explanations (passive flows, sector composition, earnings growth of the largest firms), each with one implication the others do not share
 - ▶ **data and code**: SPY and RSP adjusted (total-return) prices, joined on common days; first replicate SPY $\times 12.7$ and RSP $\times 11.9$
- ▣ Test: $d_t = r_{\text{SPY},t} - r_{\text{RSP},t}$ (daily log returns); $H_0 : \mathbb{E}[d_t] = 0$ against $H_1 : \mathbb{E}[d_t] \neq 0$
- ▣ Checks fixed in advance: the price type, the calendar join, the windows (full sample and two halves) and the HAC (heteroskedasticity and autocorrelation consistent) Newey–West t -statistic

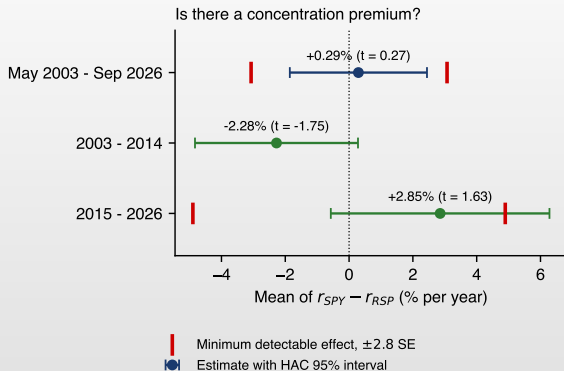
Estimation and Power

- May 2003 – September 2026 (5 884 days): mean spread +0.29% a year, $t = 0.27$
- Pre-specified halves: 2003–2014 −2.28% a year ($t = -1.75$); 2015–2026 +2.85% ($t = 1.63$)
- **Minimum detectable effect (MDE)**: the smallest true effect found with 80% power at the 5% level

$$\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.8 \times \text{SE}$$

- ▶ full sample: 3.1% a year; 2015–2026: 4.9% a year, larger than the estimate itself
- “Not significant” is not “no effect”: here the design cannot detect premiums below about 3% a year
- Causality: SPY – RSP is an outcome, not a measure of passive ownership; a causal claim needs an exogenous shifter

Mini-Case: Estimate, Interval and MDE



- Dots: annualised mean of d_t ; bars: HAC 95% intervals; red ticks: $\pm$ MDE
- Every interval contains 0
- The sign flips between the two halves: the “premium” is a feature of the chosen window
- Report every window tried, not only the one that looks significant

Your Turn: A Project Seed

- ▣ **Question:** does the SPY – RSP spread survive after controlling for sector returns?
 - ▶ regress d_t on the sector ETFs (XLK, XLC, XLF, ...) from the course market data; test H_0 : intercept = 0 with HAC standard errors
- ▣ **Replicate first:** SPY $\times 12.7$, RSP $\times 11.9$ and the mini-case t -statistics
- ▣ **Pre-register:** H_0 , H_1 , test, sample and corrections, committed before the results
- ▣ **Report:** estimate, standard error, MDE and every specification tried; the code as a Quantlet
- ▣ **Log:** prompts that shaped decisions in AI_USE.md, every caught error in AI_ERRORS.md; reusable prompts for each step are in the [▶ Appendix](#)

Key Takeaways

- ▣ Markets in 2026 combine exchanges, OTC dealers, ETFs, algorithms and blockchains; the observation time of a “daily price” depends on the venue
- ▣ From the Amsterdam exchange (1602) to machine learning and quantum computing, each wave of models came with new data and new computing power; bubbles and crashes have recurred for four centuries
- ▣ 2000–2026 repeats the same patterns: deep drawdowns with slow recoveries, short volatility spikes, correlations that change with the macro regime
- ▣ Every number is an estimate
 - ▶ standard error of an annualised mean return: $\sigma_a/\sqrt{Y}$; of an annualised Sharpe ratio: about $1/\sqrt{Y}$
 - ▶ $\sqrt{h}$ scaling, correlations and drawdowns rest on assumptions that the data often violate
- ▣ Cap-weighted passive investing carries market concentration; crypto moved closer to mainstream finance through ETFs and stablecoins
- ▣ Data choices (field, calendar, source) are modelling decisions

Check Yourself, and Next: Chapter 1

Retrieval questions

- Why does the SE of a mean return not fall when you use daily instead of monthly data?
- Why must prices be joined on common days before computing Bitcoin – S&P 500 returns?
- Why is the SPY/RSP ratio not a measure of passive ownership?

: stylised facts

- How heavy are the tails of daily returns, and is the kurtosis even finite?
- Does volatility have long memory?
- At which horizon do returns approach the Normal distribution?
- Do the facts hold for the S&P 500, Bitcoin, BVB stocks and EUR/RON?

Appendix: Course Map, 20 Chapters

- ▣ **0** Financial Markets in 2026
- ▣ **1** Stylised Facts and Returns
- ▣ **2** Market Efficiency: From EMH to Adaptive Markets
- ▣ **3** Factor Models and Asset Pricing
- ▣ **4** Portfolio Optimisation
- ▣ **5** Conditional Volatility: GARCH Models
- ▣ **6** Multivariate Volatility and Dependence
- ▣ **7** Value-at-Risk and Expected Shortfall
- ▣ **8** Backtesting and Evaluating Risk Forecasts
- ▣ **9** Realised Volatility
- ▣ **10** Market Microstructure
- ▣ **11** Continuous-Time Models
- ▣ **12** Options and the Volatility Surface
- ▣ **13** Machine Learning in Finance
- ▣ **14** Deep Learning and Time-Series Foundation Models
- ▣ **15** LLMs and Sentiment Analysis
- ▣ **16** Digital Assets and DeFi
- ▣ **17** Bubbles and Crashes
- ▣ **18** Systemic Risk and Networks
- ▣ **19** Review and Project Presentations



Appendix: Core Reading and Further Study

- ▣ Franke, Härdle & Hafner (2019), *Statistics of Financial Markets*, 5th ed.: the main statistical reference
- ▣ Campbell, Lo & MacKinlay (1997), *The Econometrics of Financial Markets*; Brandimarte (2018), *An Introduction to Financial Markets: A Quantitative Approach*
- ▣ Härdle & Simar (2019), *Applied Multivariate Statistical Analysis*; Härdle, Chen & Overbeck (2017) (Eds.), *Applied Quantitative Finance*, 3rd ed.
- ▣ Härdle, Harvey & Reule (2020), "Understanding cryptocurrencies", *Journal of Financial Econometrics*
- ▣ Advanced courses on quantinar.com
 - ▶  Statistics of Financial Markets;  Digital Transformation on Finance
 - ▶  Cryptos, NFTs, Digital Assets;  Introduction to Blockchain and Cryptocurrencies

Appendix: Asset-Class Terms

- ▣ **T-bills:** Treasury bills, government debt with maturity below one year; **repo:** repurchase agreement, a short-term loan backed by securities
- ▣ **Futures:** a contract to buy or sell later at a price fixed today; **roll yield:** the gain or loss from replacing an expiring contract with the next one
- ▣ **Options:** the right, not the obligation, to buy or sell at a fixed price; **swaps:** exchanges of cash-flow streams
- ▣ **Idiosyncratic risk:** specific to one company; **counterparty risk:** the other side of a contract may not pay
- ▣ **Multilateral trading facility (MTF):** a trading venue run outside a regulated exchange; **central counterparty (CCP):** becomes the buyer to every seller and the seller to every buyer; **custodian:** keeps the securities

Appendix: From the Trading Floor to Screens



NYSE trading floor, 2011

Photo: Carol M. Highsmith (2011);
public domain; Wikimedia Commons



Electronic displays, Chicago Board of Trade, 1992

Photo: Lars Plougmann (1992); CC BY-SA 2.0;
Wikimedia Commons

- **Open outcry:** traders met on the floor and matched orders by voice and hand signals
- Electronic limit order books gradually replaced most floor trading
- Today most volume is matched by computers (market microstructure, [Chapter 10](#))

Appendix: A Historic Crash, Black Tuesday, 1929



Crowd outside the NYSE, 29 October 1929

Photo: U.S. government (1929); public domain;
[Wikimedia Commons](#)

- 29 October 1929 (**Black Tuesday**): crowds gather outside the New York Stock Exchange during the crash
- The crash of October 1929 deepened a US recession that had begun in August 1929; the initial contraction of the Great Depression lasted until 1933 ([NBER business-cycle dates](#))
- Long, deep drawdowns are not new; bubbles and crashes are modelled in [Chapter 17](#)

Appendix: The Roll Model of Bid–Ask Bounce

- Model: $p_t = m_t + (s/2) q_t$, with m_t the efficient price (a random walk), s a constant spread in price units, $q_t = \pm 1$ independent, equally likely trade signs, unrelated to Δm_t
- Then $\Delta p_t = \Delta m_t + (s/2)(q_t - q_{t-1})$ and only the shared q_{t-1} links consecutive changes:

$$\text{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{s^2}{4} \text{Var}(q_{t-1}) = -\frac{s^2}{4}, \quad \hat{s} = 2\sqrt{-\hat{\gamma}_1}$$

- If $\hat{\gamma}_1 > 0$ the estimator is undefined ($\hat{\gamma}_1 = 0$ gives $\hat{s} = 0$); other dependence also moves $\hat{\gamma}_1$, so not every negative autocovariance is bounce ([Roll \(1984\)](#); [Chapter 10](#))

Appendix: Variance Ratios and Variance Estimates, More Examples

- EUR/RON, BNR reference rate, 2005–2026: $\hat{\rho}_1 = 0.17$
 - ▶ a hypothetical AR(1) would give the factor $\sqrt{1.17/0.83} = 1.19$ (the 2005–2026 volatility annualised with $\sqrt{252}$, 4.4%, would become 5.3%)
 - ▶ but $\hat{\rho}_2, \hat{\rho}_3, \hat{\rho}_4 < 0$ offset lag 1: monthly returns give 4.4% (Seminar 0, A6)
- EUR/RON series from EODHD: bad ticks give $\hat{\rho}_1 = -0.57$ and inflate $\sqrt{252}$ volatility; they are errors to remove, not an AR(1)
- Variance behaves differently from the mean: $\text{Var}(\hat{\sigma}^2) \approx \sigma^4(\kappa - 1)/T$ shrinks with T , slowly when the kurtosis κ is large (SPY daily: $\kappa = 17.1$, [Chapter 1](#))

Appendix: The Sharpe Ratio under Fat Tails, Dependence and Selection

- Non-Normal iid returns with skewness γ_3 and kurtosis γ_4 : the asymptotic variance of $\sqrt{T}(\widehat{\text{SR}}_d - \text{SR}_d)$ is

$$V = 1 - \text{SR}_d \gamma_3 + \text{SR}_d^2 (\gamma_4 - 1)/4, \quad \text{SE}(\widehat{\text{SR}}_d) = \sqrt{qV/T}$$

- ▶ Opdyke (2007); SPY: SE 0.30 instead of 0.29, because the daily SR is small
- Dependent returns: HAC or studentised bootstrap inference (Ledoit & Wolf (2008))
- Selection: picking the best of many assets or strategies ex post inflates the SR; the deflated SR corrects for it (Bailey & López de Prado (2014), Chapter 13)

Appendix: Asynchronous Closes

- Bitcoin's price for day t is taken at 24:00 UTC; the S&P 500 closes at 20:00 or 21:00 UTC: news arriving between the two times enters the two series on different days
- With a common factor, the same-day correlation is attenuated and the rest appears in lead/lag correlations
 - ▶ if only the common factor is split across days and daily variances are unchanged, the lead, same-day and lag correlations sum to the true one (Scholes & Williams (1977); Lo & MacKinlay (1990); Seminar 0, A9)
 - ▶ otherwise use longer returns (Burns, Engle & Mezrich (1998))
- Bitcoin – S&P 500, 2022–2026, common days: $0.42 + 0.03 - 0.05 = 0.40$; the sum is a diagnostic, not a proof

Appendix: Correlation in Turbulent Periods, the Volatility Bias

- Stable link $y = \beta x + \varepsilon$ (β and $\text{Var} \varepsilon$ fixed) with correlation ρ ; estimated only on days when $\text{Var}(x)$, the source market, is $(1 + \delta)$ times higher, it gives ([Forbes & Rigobon \(2002\)](#))

$$\rho^* = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}}, \quad |\rho^*| > |\rho| \text{ for } \delta > 0, \quad 0 < |\rho| < 1$$

- Bitcoin – S&P 500, 2022–2026, days split by the VIX (top quartile: $\text{VIX} > 21.8$)
 - ▶ 0.52 on high-VIX days vs 0.32 on the others; S&P 500 variance ratio $1 + \delta = 4.64$
 - ▶ adjusted $\hat{\rho} = \rho^* / \sqrt{1 + \delta(1 - \rho^{*2})} = 0.27$
- The adjustment assumes no omitted common shock and no feedback between the two markets

Appendix: Selection, Survivorship and Horizon Effects

- ▣ **Survivorship:** samples rebuilt from surviving firms and coins make past returns look too good (Brown, Goetzmann, Ibbotson & Ross (1992))
- ▣ **Selection:** Bitcoin's $\times 257$ is chosen ex post among thousands of coins; a best-of-many Sharpe ratio needs a deflated SR
- ▣ **Horizon:** the depth of the maximum drawdown grows with the sample length
 - ▶ log price a driftless Brownian motion: the log drawdown $D = \max_{s \leq t \leq T} (\ln P_s - \ln P_t) \geq 0$ has $E[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ (Magdon-Ismail et al. (2004))
 - ▶ $\sigma = 17.6\%$: $E[D] = 0.70$ over 10 years, 1.40 over 40; as price declines $e^{-E[D]} - 1 = -50\%$ and -75%
 - ▶ the expected percentage MDD, $E[e^{-D}] - 1$, is smaller in magnitude (Jensen): about -47% and -71% in simulated paths
- ▣ The BET's -82.5% (29 years) and an MDD from 11.7 years of data are not comparable without the same T

Appendix: ETF Premium Dynamics

- Statistical view of the arbitrage: an AR(1) premium

$$\pi_t = \phi \pi_{t-1} + u_t, \quad \text{half-life} = \frac{\ln 0.5}{\ln \phi}, \quad 0 < \phi < 1$$

- ϕ close to 1: premiums decay slowly, as for ETFs on foreign stocks or illiquid bonds ([Petajisto \(2017\)](#))
- In the US, in-kind redemptions also let the fund avoid distributing capital gains (a US tax rule, not a general feature)

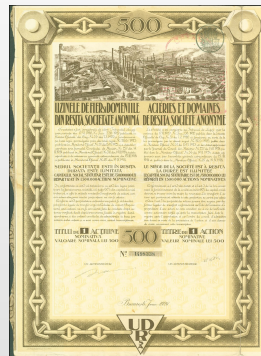
Appendix: Regulation of Digital Assets

	European Union	United States
Framework	MiCA, Regulation (EU) 2023/1114	GENIUS Act, Public Law 119-27 for payment stablecoins
Crypto ETPs	exchange-traded notes on crypto	spot Bitcoin ETPs approved in January 2024 (SEC (2024))
Supervisors	ESMA, EBA, national authorities	SEC, CFTC, banking regulators

- ESMA: European Securities and Markets Authority; EBA: European Banking Authority; SEC: Securities and Exchange Commission; CFTC: Commodity Futures Trading Commission
- BIS (Bank for International Settlements, the bank of central banks): BIS (2023) propose a “unified ledger” combining central bank money, deposits and assets; BIS (2025): stablecoins fall short of the three tests for money, **singleness** (accepted at par by all), **elasticity** (supply follows payment needs) and **integrity** (resists financial crime)

Appendix: The Bucharest Stock Exchange, 1882–1948

- **1 Dec 1882:** the securities and commodities exchange opens under a royal decree of King Carol I [BVB \(history\)](#)
- Early shocks [BVB press release \(2017\)](#)
 - ▶ the 1907 peasants' revolt and the Balkan wars of 1912–1913
 - ▶ closed during the First World War
- After reopening: about 7 years of strong growth, then about 7 years of accelerated decline, overlapping the Great Depression of 1929–1933 [BVB press release \(2017\)](#)
- **1929:** a new law on exchanges, with the explanatory memorandum of minister Virgil Madgearu [Lege asupra burselor \(1929\)](#)
- **1948:** trading stops with the communist regime and the nationalisation of the economy [BVB \(history\)](#)



Share of UDR (Uzinele și Domeniile Reșița), June

1926. Scan: EDHAC e.V.; public domain;

[Wikimedia Commons](#)

[← Back](#)

Appendix: Palatul Bursei, the Exchange's Historic Home



Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons



Photo: Chainwit. (2023); CC BY 4.0; Wikimedia Commons

- Built 1906–1912, architect Ștefan Burcuș, Str. Ion Ghica 4 [Palatul Bursei](#)
- Seat of the Exchange and of the Bucharest Chamber of Commerce and Industry
- From 1955: the State Central Library (today the National Library)
- Left: stockbrokers at the entrance, March 1928; right: the building in 2023

Appendix: Interruption and Rebirth, 1948–2023

- ▣ **1948–1995:** almost 50 years without a stock exchange [BVB press release \(2017\)](#)
- ▣ **20 Nov 1995:** first session of the re-established exchange: 905 shares of 6 listed companies [BVB press release \(2017\)](#)
- ▣ **1997:** launch of BET on 19 Sep 1997 at 1 000 points, with 10 companies; the basket may hold 10–20 and today has 20 [BVB \(indices\)](#)
 - ▶ **BET-FI:** the first sector index, for the financial investment companies (SIFs, listed in 1999)
- ▣ **2010:** BVB lists its own shares on its **regulated market** (an EU-authorized trading venue) [BVB \(history\)](#)
- ▣ **Sep 2020:** **FTSE Russell**, a global index provider, reclassifies Romania from Frontier to (Secondary) Emerging [BVB \(history\)](#)
- ▣ **Jul 2023:** Hidroelectrica lists through the largest IPO in BVB history, EUR 1.9 bn [BVB \(history\)](#)
- ▣ Largest daily BET move: 7 Jan 2009, −12.3% (−13.1% log), after +4.8% the day before (official BVB daily archive [BVB \(indices\)](#))

Appendix: Case Study, Data, Demand System and Estimation

- **Data:** quarterly Form 13F holdings, filed with the SEC by institutional managers with discretion over at least USD 100 million in 13(f) securities, 2001–2020; 80% of US market capitalisation (Section III.B)
 - ▶ characteristics X_k : log book equity, profitability, investment and dividend yield, each transformed to follow the Normal distribution, plus squared book equity; universe $\mathcal{N}_i$: stocks held in the past three years

- **Demand** (eq. 13–14), w_{ik} portfolio weight, 0 outside asset, p_k log market capitalisation:

$$\log(w_{ik}/w_{i0}) - p_k = \bar{d}_{0i} + \bar{d}'_{1i}X_k - \mathcal{E}_{ik} p_k + \xi \mathcal{E}_{agg,k} + \zeta' \mathcal{E}_{agg,k} X_k + \epsilon_{ik}$$

- ▶ active: $\mathcal{E}_{ik} = \bar{\mathcal{E}}_{0i} + \bar{\mathcal{E}}'_{1i}X_k - \chi \mathcal{E}_{agg,k}$; passive: $\mathcal{E}_{ik} = 0$
- ▶ equilibrium (eq. 17–18): $\sum_i w_{ik} A_i = P_k$ and $\sum_i (w_{ik} A_i / P_k) \mathcal{E}_{ik} = \mathcal{E}_{agg,k}$, A_i the investor's assets
- **Instruments:** price (eq. 23) $\hat{p}_{k,i} = \log \sum_{j \neq i} A_j 1\{k \in \mathcal{N}_j\} / |\mathcal{N}_j|$; elasticity (eq. 24): the elasticity condition solved with the same equal-weighted portfolios
- **Estimator:** GMM with a nested fixed point solved by Newton's method (Section III.C); investors with fewer than 1 000 holdings pooled by AUM (assets under management) into groups of about 2 000 holdings (Table 2)



Appendix: Case Study, Complete Results

Topic	Number in the paper	Where	Reading
Strategic response	$\chi = 2.97$ (SE 0.47); $\zeta = -0.48$ (SE 0.14)	Table 2, row 1	between 0 and ∞
Robustness	χ from 2.30 to 3.27 in rows 2–14; 0.04 without the elasticity instrument, 0.63 without instruments	Table 2	endogeneity hides the response
Elasticity level	mean $\mathcal{E}_{agg}$ 0.438 (fixed elasticities 0.39); residual cross-sectional SD 0.0395 (fixed 0.0842)	Table 3	reactions halve the dispersion
Rise of passive	median active share 81.4% (2001), 59.5% (2020)	Figure 5	passive 19% to 41%
Reduced form	slope of $\Delta \log \mathcal{E}_{agg}$ on $\Delta \log$ active share 0.385 (SE 0.022); 0.336 (SE 0.055) with Russell 1000/2000 switches	Table 4, col. 2, 5	consistent with one-third
Prices	per SD of elasticity, instrumented: volatility -0.383 (SE 0.123), illiquidity -0.396 (SE 0.131)	Table 5, col. 2, 8	inelastic: more volatile
Evolution	mean $\mathcal{E}_{agg}$ 0.41 to 0.27; with $\chi = 0$ the fall about twice as large	Figures 6, 8	response damped the fall

Appendix: Case Study, Replication with Course Data

- The 13F holdings are not in the course data, so χ cannot be re-estimated; the arithmetic of Section V.A can
 - ▶ eq. (5) with $\alpha = 70\%$ active, $\chi = 3$: $0.70 + 0.30 \times 2.1/3.1 = 0.90$; eq. (28): $1/(1 + 3 \times 0.68) = 0.33$
- Our measure (ETFs only), Financial Accounts of the United States: [ETFs' corporate equities over all sectors' corporate equities](#)
 - ▶ 0.4% (2001 Q1), 6.6% (2020 Q4), 10.3% (2026 Q2)
 - ▶ changes relative to 81%: active share -7.7% , elasticity -2.5% over 2001–2020; -12.3% and -4.0% to 2026 Q2
- The paper's broader measure adds passive mutual funds: 15 points over 30 years, i.e. -19% of the 81% and -6.3% ; the Financial Accounts do not split index and active mutual funds
- The authors also point to bank regulation, fire sales and large sovereign investors (Section VI)

Appendix: A Minimal Python Workflow

```
import numpy as np, pandas as pd

URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')
def load(sym, col='close'):
    d = pd.read_csv(URL + sym + '.csv',
                    index_col='date', parse_dates=True)
    return d[col].rename(sym)

spx = load('GSPC.INDX')           # index: close
spy = load('SPY.US', 'adjusted_close') # ETF
btc = load('BTC-USD.CC')          # 24/7
r_spx = np.log(spx).diff()        # one series
# joint analysis: join PRICES on common
# days first, then take returns
p = pd.concat([spx, btc], axis=1).dropna()
r = np.log(p).diff().dropna()
dd = spx / spx.cummax() - 1       # drawdown
```

- One series: returns on its own calendar, `np.log(s).diff()`
- Joint analysis: join the *prices* on common days, then take returns, so both cover the same interval (Friday → Monday)
- Returns taken before the join would pair a Bitcoin Sunday → Monday move with an S&P 500 Friday → Monday move
- Crypto trades every day: annualise with 365, not 252

Appendix: Data Checklist

- ▣ Plot the raw series and list the largest returns
- ▣ Compare two sources when possible
- ▣ Check adjustments: dividends, splits, redenominations
- ▣ Remove spurious non-trading-day records; investigate isolated outliers, and correct or remove only confirmed data errors; document every change
- ▣ For joint analyses, join the prices on common dates before computing returns; for a single asset, use its own calendar
- ▣ Watch survivorship: delisted firms vanish from today's index lists

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ List peer-reviewed papers since 2018 on passive ownership and price informativeness. For each: DOI, identification strategy, main estimate. Flag any paper you are not sure exists.
- ▶ Resolve every DOI on Crossref; title and authors must match; read the abstract yourself

□ Hypothesis

- ▶ Give three competing explanations for SPY beating RSP after 2015 (passive flows, sector composition, earnings growth of the largest firms). For each, one testable implication the others do not share.

□ Data and code

- ▶ Write Python that reads the daily adjusted (total-return) prices of SPY and RSP from the course data, joins prices on common days, and tests H_0 : mean daily log-return spread = 0 with a Newey-West t-statistic.

Appendix: The Discovery Loop, Reusable Prompts (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: subsamples, HAC bandwidths, simple vs log returns, a sector-neutral version. State in advance which outcome would falsify the hypothesis.

▣ Critique

- ▶ Act as a hostile referee. List the three strongest threats to identification in this design and the test that addresses each.

▣ Report: estimate, standard error, sample, every specification tried, also the failed ones

▣ What the researcher must check: invented references (resolve each DOI); unadjusted instead of adjusted close (reproduce the total-return multiples first); returns before the join (count the common days); window shopping (fix the windows in advance); low power (report the MDE)

Appendix: Tulips and the South Sea Bubble, Revisited



Jan Brueghel the Younger, *Satire on Tulip Mania*, c. 1640

Painting: Jan Brueghel the Younger; public domain; [Wikimedia Commons](#)



William Hogarth, *The South Sea Scheme*, 1721

Print: William Hogarth (1721); CC0; [National Gallery of Art, Wikimedia Commons](#)

- The popular story of tulip mania comes largely from moralising pamphlets written after the crash
- [Garber \(1989\)](#): prices of rare bulbs fell the way prices of new varieties fall in later flower markets; the spike in common bulbs lasted a few weeks of winter trading in taverns, with contracts that were rarely enforced
- [Goldgar \(2007\)](#): archival records show few bankruptcies and little damage to the Dutch economy
- [Frehen, Goetzmann & Rouwenhorst \(2013\)](#): in 1720 the boom spread across Paris, London and Amsterdam; prices rose most for firms tied to Atlantic trade and to insurance
- Lesson: a bubble needs a story about a real change; dating it requires a model of fundamental value ([Chapter 17](#))

Appendix: The Gaussian Copula and 2008

- [Li \(2000\)](#): joint default times from marginal default probabilities F_i and a Normal dependence structure

$$\tau_i = F_i^{-1}(\Phi(Z_i)), \quad Z_i = \sqrt{\rho} M + \sqrt{1-\rho} \varepsilon_i, \quad M, \varepsilon_i \sim \mathcal{N}(0, 1) \text{ independent}$$

- One parameter ρ priced every tranche of a CDO (collateralised debt obligation), a security that pays from a pool of loans in order of seniority
- Weaknesses exposed in 2007–2008
 - ▶ for $\rho < 1$ the Gaussian copula has zero tail dependence: joint extreme defaults are too rare
 - ▶ the correlation implied by market prices differed across tranches, so one ρ could not fit them all
 - ▶ inputs were estimated over calm years of rising house prices
- Lesson: model risk; dependence in the tails ([Chapter 6](#)) and systemic risk ([Chapter 18](#))

Appendix: Classical Monte Carlo versus Quantum Amplitude Estimation

- ▣ Classical MC: $\hat{\mu}_N = \frac{1}{N} \sum_{j=1}^N f(X_j)$ has standard error $\sigma/\sqrt{N}$, so accuracy ε needs $N = O(\sigma^2/\varepsilon^2)$ samples
- ▣ QAE writes μ into the amplitude of a qubit, $\sqrt{1-\mu}|0\rangle + \sqrt{\mu}|1\rangle$ (for f scaled to $[0, 1]$), and applies a Grover-type operator m times; the error falls as $O(1/m)$
- ▣ Hence $O(1/\varepsilon)$ calls of the pricing circuit ([Montanaro \(2015\)](#); [Rebentrost, Gupt & Bromley \(2018\)](#))
- ▣ *Quadratic* means: where classical MC takes N steps, QAE takes about $\sqrt{N}$; an exponential gain would mean about $\log N$ steps, but for estimating a mean no quantum algorithm beats QAE ([Montanaro \(2015\)](#))
 - ▶ $\varepsilon = 10^{-4}$: about 10^8 classical samples against about 10^4 quantum calls
- ▣ Why this is not yet an advantage
 - ▶ each call must load the distribution of X and run without error, which needs error correction
 - ▶ error-corrected operations are far slower than classical ones: [Chakrabarti et al. \(2021\)](#) estimate about 8,000 logical qubits and a T-depth (sequential layers of the slowest error-corrected gate) of 54 million
 - ▶ classical MC parallelises easily across many processors

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